

HW3: Rocky Times Challenge - Search, Map and Analyze

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1 Introduction

The objective of this assignment was to design an autonomous drone system capable of exploring an unknown environment, detecting cylindrical structures, estimating their dimensions, and performing precision landing on the tallest cylinder using ArUco markers.

The system was implemented using PX4 SITL, Gazebo, and ROS2 Humble, integrating perception, control, and decision-making into a complete autonomous pipeline.

2 System Setup

The environment was set up using PX4 SITL and ROS2. The simulation was launched with the custom drone model:

```
cd ~/PX4-Autopilot
make px4_sitl gz_x500_depth_mono
```

The ROS2 workspace included the terrain mapping package, and communication with PX4 was established using:

```
MicroXRCEAgent udp4 -p 8888
```

The mission was executed using:

```
ros2 launch terrain_mapping_drone_control mission.launch.py
```

3 Autonomous Mission Flow

The mission logic was implemented as a state machine in the `cylinder_mission_node`. The major states are:

- **WAIT_INTRINSICS**: Waits for camera and battery data
- **ARM_TAKEOFF**: Arms and takes off to a fixed height
- **CIRCLE**: Follows a predefined circular trajectory
- **SERVO**: Moves toward detected cylinder
- **HOVER**: Stabilizes and measures object dimensions

- **ARUCO_HOVER:** Moves above cylinder for marker detection
- **ARUCO_SELECT:** Selects the best marker (tallest cylinder)
- **ARUCO_MOVE:** Aligns with marker
- **LAND:** Executes precision landing

The circular path followed by the drone is predefined using:

$$x = r \cos(\theta), \quad y = r \sin(\theta), \quad z = \text{constant}$$

4 Cylinder Detection Pipeline

Cylinder detection is performed using RGB-D data:

4.1 Steps

1. Convert RGB to HSV and apply thresholding
2. Apply depth filtering (1m to 20m)
3. Combine masks and apply morphological operations
4. Extract contours and filter by area
5. Estimate real-world dimensions:

$$\text{Width} = \frac{w \cdot Z}{f_x}, \quad \text{Height} = \frac{h \cdot Z}{f_y}$$

This approach provides robust detection, although noise and lighting introduce small errors.

5 Target Selection Logic

The drone stores dimensions of detected cylinders and compares them.

If a cylinder is detected multiple times with matching dimensions, it is identified as the target. The tallest cylinder is selected for landing.

6 ArUco-Based Landing

ArUco markers are used for final precision landing. Marker positions are obtained from the `/aruco/marker_pose` topic.

The transformation used is:

$$x_{\text{drone}} = y_{\text{aruco}}, \quad y_{\text{drone}} = x_{\text{aruco}}, \quad z_{\text{drone}} = z_{\text{aruco}}$$

The drone selects the marker with minimum Z (closest) and moves toward it using proportional control.

7 Results

The system successfully completed the mission:

- Multiple cylinders detected and measured
- Tallest cylinder correctly identified
- ArUco markers successfully tracked
- Precision landing achieved

7.1 Visual Results

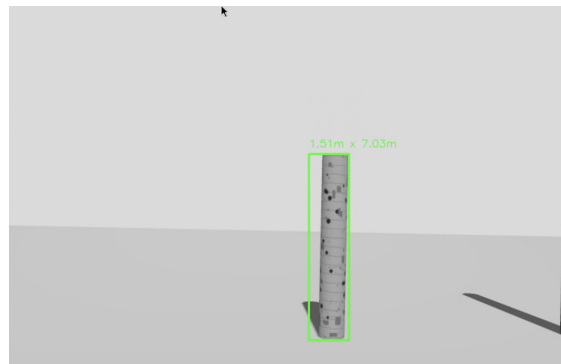


Figure 1: Short Cylinder Detection

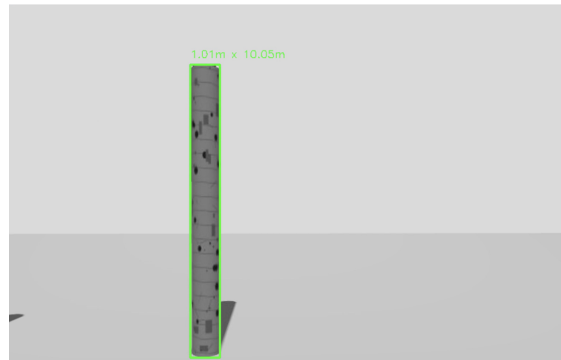


Figure 2: Tall Cylinder Detection

7.2 Performance Metrics

- Mission Duration: 177.91 seconds
- Battery Used: 0% (simulation artifact)

8 Challenges Faced

Several challenges were encountered during development:

- **PX4 Connection Issues:** The system initially failed to connect to the ground control station, preventing arming.
- **Offboard Mode Activation:** The drone did not respond to commands due to incorrect offboard initialization.
- **Aruco Detection Failure:** The major issue was that `len(self.marker) = 0`, meaning no markers were detected. This was caused by:
 - Incorrect camera topic selection
 - Camera not facing downward during hover
 - Marker not visible within field of view
- **Hover Alignment Issues:** The drone hovered at incorrect positions, preventing proper marker detection.
- **ROS2 Topic Debugging:** Topics such as `/aruco/marker_pose` were not publishing initially, requiring debugging of the detection node.

These issues were resolved by fixing camera alignment, verifying ROS topics, and ensuring correct state transitions.

9 Demonstration Video

A full demonstration of the mission, including autonomous navigation, detection, and landing, can be found at:

Mission Demo Video

10 Conclusion

A complete autonomous drone system was successfully implemented, integrating perception, control, and decision-making.

The drone was able to explore, detect cylinders, identify the tallest structure, and perform a precise ArUco-based landing.

11 Future Work

- Replace predefined circular motion with SLAM-based exploration
- Improve robustness of ArUco detection
- Incorporate better sensor fusion
- Improve energy estimation beyond simulation limitations